

Plantim Icons v4.1 — proposal report

New icons for loading states and placeholders, the specific cases the app currently works around, and a country-flag asset class — authored in the v4 design language, gated mechanically, and delivered as a review package (interactive decision catalog + PDF catalog + this report). Additive only: no shipped id changes.

41	27	14	816
PROPOSED IDS	SEMANTIC ICONS	FLAGS	GENERATED SVGS



proposalHash 520b57ace477bc3314f4ebf9efa9b9cb8ae8bac5f9a43e3d88fb49b1a98a4071
derived from v4 4.0.0 · v4Hash f12d51dd066e0493bb460f13ca0d9c957e58542320303fbce9aadf893972be4b

Summary

The v4 rework (237 icons) covers the app's existing icon surface. This proposal fills three gaps the app audit (reports/v4-app-icon-audit.md) exposed but v4 deliberately left out of scope:

- 1. **Loading states & placeholders** (9 ids) — the app ships profile-placeholder.svg as a raw asset, scales plant.sprout to 56 px as the plant-photo fallback and has no skeleton, pending or offline glyphs at all.
- 2. **Specific cases** (18 ids) — ids with no semantic equivalent in v2/v3/v4 that the clients fake with a neighbouring glyph (download via upload, room via navigation.home, add-photo via camera, add-event via action.add, humidity via the watering drop, light via the weather sun, ...).
- 3. **Country flags** (14 ids) — a language/region picker needs them; the web client already carries assets/flags/*.svg outside the registry, so iOS and Android would each re-source them.

Everything is authored in the v4 DSL (same grid, strokes, grades, layer roles, color tokens) and passes the same gates, so an approved id promotes into v4 by moving its module — no redraw, no adapter change beyond regeneration. Flags use a dedicated, narrower DSL and renderer because they are not monochrome semantic icons (see §3).

Method

- **Grounding.** Every id cites the surface that needs it (audit §2 usage table, §3 gaps, §3c hardcoded assets). Ids that an existing semantic id already covers were *not* proposed (e.g. a generic image placeholder is utility.image at ink.300; a generic empty state is utility.inbox).
- **Authoring.** 5 hand-authored modules under design-tokens/icons/v4.1-proposal/src/ using the v4 DSL (icon / layer / P / C / R): 24-grid, 2 px round strokes, named layers with hand-assigned roles, per-icon multicolor tokens, declared solid counters. 10 of 27 icons carry an authored micro grade; 8 have true-fill solids with declared counters, the rest use bold-mode solids (line-only geometry, the v4 P1 convention).
- **Gates.** check-proposal-icons.mjs runs the v4 structural checks (hash integrity, determinism via --verify, source hygiene, path-walked bounds and 3-decimal clamp, per-grade stroke discipline, 4 × 6 size-set completeness, a11y keys, token refs) plus proposal rules (ids must be new across v2/v3/v4; tiers P1/P2 only; every status/semantic label present in en/de/tr) and flag rules (clip + border present, nocurrentColor in color variants, canonical hex, micro survivors). --raster adds the v4 flood-fill counter check and outline ↔ solid silhouette IoU ≥ 0.90 for the 8 filled solids. Both pass on this build.
- **Human loop.** Contact sheets (render-v4-sheets.mjs proposal: four variants at 128 px, 16 → 72 ramp, solid + multicolor on dark, 16–24 on dark) inspected zoomed; two icons reworked after the first pass (plant.light rays converged into an arrowhead → now radiate from the corner; care.misting trigger lengthened), one skeleton solid re-proportioned to pass the silhouette gate.

1. Loading states & placeholders 9 ids

Skeleton and fallback glyphs for the moments before content exists: profile photo, plant photo, list rows, cards, charts, plus inline loading, pending and offline states. Today these are raw assets (profile-placeholder.svg), scaled-up tab icons (plant.sprout at 56 px as the plant-hero fallback) or nothing at all.

Design rule. Container + ghost content. Multicolor stays inside the ink ramp (ink.300 container, ink.500 content) so a placeholder recedes; duotone mutes the container. Solids are real tiles, so a placeholder can occupy exactly the footprint of the asset it stands in for.

24 PX · 16 PX	ID · WHY · WHERE
	placeholder.avatar (P1) Ring + bust with the shoulders meeting the ring — the classic 'no photo' crop. Distinct from account.user (Profile tab) at every size; the solid keeps the space around the bust as its counter. web: assets/profile-placeholder.svg (raw asset, outside the registry) · ios: ProfileView falls back to account.user · member lists, comments, public profile
	placeholder.card (P1) Media block + two lines: card skeleton for plant / garden grids and the social feed. Garden grid, Plants grid, SocialFeed cards
	placeholder.chart (P2) Axes + ghost bars for chart loading/empty. utility.bar.chart3 stays the 'chart' action glyph. web: GrowthChart.vue empty state (utility.bar.chart3 h-7 today) · stats tiles

	<code>placeholder.list</code> P1 Avatar disc + two lines: a list-row skeleton (notifications, members, followers). NotificationFeed, InviteMemberSheet, garden members
	<code>placeholder.plant</code> P1 Photo tile + leaf. Replaces the scaled-up plant.sprout used as the plant-hero fallback; solid = filled tile with the leaf as counter. web: PlantDetailsHero.vue:357 (plant.sprout h-14) · plant cards without a photo · AddPlant preview
	<code>placeholder.text</code> P1 Two full lines + one short — the canonical text skeleton. Bold solid gives fat bars for shimmer overlays. feed / notification / journal rows while loading · Today card summaries
	<code>status.loading.dots</code> P1 Bouncing-dots snapshot (middle dot raised) so it never collides with utility.more.horizontal; multicolor/duotone carry the trail. Reuses the existing a11y.icons.status.loading label. Chat: 'AI is typing' · inline saves, optimistic updates · static fallback where Lottie is unavailable status: Loading
	<code>status.offline</code> P1 Cloud + slash. The mobile client has no connectivity glyph; the status color lands on the slash. offline banner (iOS/PWA) · sync indicator in Settings status: Offline
	<code>status.pending</code> P1 Hourglass for waiting states that are not spinners: pending invites, queued sync, unverified email. InviteMemberSheet, garden co-owner invites · VerifyEmail · offline queue status: Pending

2. Specific cases the app works around 18 ids

Ids with no semantic equivalent in v2/v3/v4 that the clients currently fake with a neighbouring glyph: download (upload exists), sort, drag handle, QR, external link, language picker, add-photo, add-event, plant identification, humidity/light requirements, pet safety, misting/rotating care types, and room/indoor/outdoor placement (navigation.home doubles as "room" on both platforms).

Design rule. Same v4 DSL, gates and grades as the shipped set. Siblings reuse v4 geometry verbatim where a pair must match (calendar.add ← calendar.date frame; utility.camera.plus ← utility.camera body; utility.download ← utility.upload tray).

24 PX · 16 PX	ID · WHY · WHERE
	<code>calendar.add</code> P1 calendar.date's hand-drawn frame + rings with a plus in the body; the plus survives the solid as a counter. web: AddCareModal.vue, CalendarSidebar.vue (action.add today) · ios: AddCareModalView.swift
	<code>care.misting</code> P2 Spray bottle + mist dots. Completes v3's care.watering/fertilizing/pruning/repotting with the one care type they lack. care calendar categories · care rhythm strip
	<code>care.rotating</code> P2 Pot + clockwise arrow: turn the plant toward the light (even growth). care calendar categories · care suggestions
	<code>location.indoor</code> P2 Window (frame + mullions) on a sill — distinct from utility.grid2x2. placement attribute on plants / gardens
	<code>location.outdoor</code> P2 Single round-crown tree on the ground; utility.trees is two pines. placement attribute on plants / gardens · balcony / garden gardens

	<code>location.room</code> P1 Door on a floor line; the knob survives the solid as a counter. Frees navigation.home, which currently means both 'Today' and 'room' (audit §3a-5). web/ios: RoomSelect, room chips · RoomScene / RoomColumn headers
	<code>plant.humidity</code> P1 Two drops — never confused with the single plant.watering drop (47 refs). plant detail environment row · weather widgets / LocationPostcard
	<code>plant.identify</code> P1 Viewfinder brackets + leaf: AI plant identification. Today the flow borrows utility.sparkles + utility.camera. AddPlant identify step · Chat plant-picker overlay
	<code>plant.light</code> P1 Leaf lit by three rays from the corner: light requirement. weather.sun (17 refs) currently doubles as theme toggle and weather. plant detail environment row · species select · room light hint
	<code>plant.pet.safe</code> P2 Paw in leaf green. Pet safety is the most-asked species attribute after light and water. species attributes · plant detail badges semantic: Pet safe
	<code>plant.pet.unsafe</code> P2 Same paw with a slash; status.error on the slash in multicolor. Semantic pair with plant.pet.safe. species attributes · plant detail warning badge semantic: Toxic to pets
	<code>utility.camera.plus</code> P1 The v4 camera body with a plus in the lens: 'add photo' as a distinct CTA. utility.camera (24 refs) currently means both 'photo' and 'add photo'. profile photo · plant photo · journal entry
	<code>utility.download</code> P1 Mirror of utility.upload (same tray, reversed arrow). The registry has upload/upload.cloud but no download. privacy: export my data · save plant photo · download care report / calendar
	<code>utility.drag.handle</code> P2 2×3 grip for reordering; utility.move (4-way arrows) means relocate, not reorder. reorder rooms, plants in a garden, care steps
	<code>utility.external.link</code> P1 Box + arrow leaving top-right — the universal 'opens elsewhere' affordance the policy pages lack. PrivacyPolicy, AccessibilityStatement links · species source links
	<code>utility.language</code> P1 Speech bubble + 'A': the language/region picker trigger. Pairs with the flag.* asset class; crossbar dropped at micro. Settings: language & region · onboarding locale step
	<code>utility.qr.code</code> P2 Three finder patterns + modules; utility.scan.line stays the scanning action. True-fill solid with the finder rings as counters. share garden / public profile via QR · scan plant label
	<code>utility.sort</code> P1 Paired up/down arrows for list ordering. utility chevrons.up.down is the select affordance, not sort. Plants list, Calendar, Feed ordering toolbar

3. Country flags 14 ids

A locale/region picker needs flags; the web client already ships assets/flags/*.svg outside the registry. Proposed as a new asset class `flag.\*` — not semantic icons — with a `color` variant (fixed vexillological colors, clipped 20×14 rounded rect, 12 % hairline border) and a `mono` variant (frame + partition lines in currentColor) for tinted contexts.

Design rule. Starter set = the three shipped locales (de, en → gb/us, tr), the EU, nine likely-next European regions, and a neutral `flag.unknown`. Icon-size simplification is deliberate (no coats of arms, 7 stripes / 8 stars on US, single-layer saltire on GB): a 16 px flag reads by silhouette and color. Colors are never theme tokens; dark mode is handled by the border, not by re-coloring.

24 PX · 16 PX	ID · WHY · WHERE
	flag.at P1 likely next region for a European plant-care audience Settings: language & region · LocationPostcard semantic: Austria
	flag.ch P1 likely next region for a European plant-care audience Settings: language & region · LocationPostcard semantic: Switzerland
	flag.de P1 shipped locale (de) Settings: language & region · profile LocationPostcard · public profile semantic: Germany
	flag.es P1 likely next region for a European plant-care audience Settings: language & region · LocationPostcard semantic: Spain
	flag.eu P1 region grouping / GDPR surfaces privacy & data region semantic: European Union
	flag.fr P1 likely next region for a European plant-care audience Settings: language & region · LocationPostcard semantic: France
	flag.gb P1 shipped locale (en) Settings: language & region · profile LocationPostcard · public profile semantic: United Kingdom
	flag.it P1 likely next region for a European plant-care audience Settings: language & region · LocationPostcard semantic: Italy
	flag.nl P1 likely next region for a European plant-care audience Settings: language & region · LocationPostcard semantic: Netherlands
	flag.pl P1 likely next region for a European plant-care audience Settings: language & region · LocationPostcard semantic: Poland
	flag.pt P1 likely next region for a European plant-care audience Settings: language & region · LocationPostcard semantic: Portugal

 	<code>flag.tr</code> P1 shipped locale (tr) Settings: language & region · profile LocationPostcard · public profile semantic: Türkiye
 	<code>flag.unknown</code> P2 neutral fallback when no region is set (never an empty slot) Settings · public profile semantic: Region not set
 	<code>flag.us</code> P1 shipped locale (en) Settings: language & region · profile LocationPostcard · public profile semantic: United States

Flags are an asset class, not semantic icons

- **Why not the v4 registry.** v4 guarantees four variants sharing one silhouette, `currentColor` strokes and theme tokens. A flag has none of those properties: its colors are fixed by vexillology, an "outline flag" is meaningless, and it must not re-theme in dark mode. Forcing flags through the v4 gates would either weaken the gates or produce wrong flags.
- **Two variants.** `color` — the canonical flag, clipped to a 20×14 rounded rect (rx 2) at (2,5) on the 24-grid, with a 12 % black hairline inset border so white edges (PL, AT, NL) keep a silhouette on light UI and saturated fields stay soft on dark UI. `mono` — frame + partition lines in `currentColor` for tinted lists, inactive states and monochrome contexts, at the v4 grade stroke (2.4 / 2 / 1.6).
- **Sizes.** The same six sizes and three grades as v4; `micro` (16/20) drops layers flagged `micro: false` (US stars, GB saltire, PT shield) and grows the EU stars so the ring closes.
- **Simplification policy.** No coats of arms (ES), 7 stripes + 8 stars (US), single-layer saltire (GB), armillary sphere as three discs (PT). The goal is recognition at 16 px, not heraldic accuracy — the policy every icon-size flag set follows.
- **Colors.** Commonly published values, uppercase hex, authored per flag; never tokens. The structural gate rejects `currentColor` in a color variant and non-canonical hex anywhere.
- **Accessibility.** Every flag is semantic with a localized label key (`ally.icons.flag.<code>`) in `en/de/tr`, so a language picker announces "Deutschland", not "flag".
- **Scope.** 14 flags: the three shipped locales (`de`, `en` → `gb` + `us`, `tr`), the EU, nine likely-next European regions, and `flag.unknown` for "region not set". Adding a flag is one `flag()` call; the id space is ISO 3166-1 alpha-2.

Accessibility

- **Status icons** (`status.loading.dots`, `status.offline`, `status.pending`) require a label in every locale; `status.loading.dots` reuses `ally.icons.status.loading`, the other keys live in `locales.proposal.json` and merge into `locales/*.json` on promotion.
- **Semantic icons** (`plant.pet.safe`, `plant.pet.unsafe`) and all 14 flags carry localized labels the same way.
- Placeholders are decorative: the surrounding control owns the accessible name ("Loading profile..."), and the icon must never be the only signal — the v4 status-icon rule applies unchanged.
- Placeholder multicolor stays at `ink.300/ink.500`: it is meant to recede. Where a placeholder is the only content on screen, pair it with text.

Platform delivery (on promotion)

- **Web.** Semantic icons flow through the existing `@plantim/icons/v4` subpath (count-agnostic generator: no code change). Flags need a sibling subpath `@plantim/icons/flags` exporting `PlantimFlag` (props: `code`, `variant = color | mono`, `size`) plus tree-shakable per-flag modules; the generator is a ~1:1 copy of `generate-v4-package.mjs` over `registry.flags`.
- **iOS.** Semantic icons: `PlantimIconV4` regeneration. Flags: a `PlantimFlag` SwiftUI view rendering the color layers as `Paths` with fixed `Color(hex:)` values (no dynamic colors) and the `mono` variant through the existing stroke renderer.
- **Android.** Semantic icons: Vector Drawables + Compose ImageVectors via `generate-v4-android.mjs`. Flags: non-tintable drawables (fixed fills, `values-night` unchanged) and a tintable `mono` drawable.
- **Size budget.** 27 icons ≈ +32 KB of per-icon modules (tree-shaken); flags ≈ +11 KB. Both stay far below the package-size gate, but the number is worth re-checking with `icons:package:size` after promotion.

Promotion checklist

1. Review in `reports/plantim-icons-v4.1-proposal-catalog.html`; export the decision record (`.md/.json`).

- 2. For each **Approve**: move the icon's `icon()` into the matching `design-tokens/icons/v4/src/<category>.mjs` (or a new module), add its tier/verdict to `tiers.json`, merge its label keys into `locales/*.json`.
- 3. Flags: land `flags/src/flags.mjs` + `lib/flag-dsl.mjs` + `lib/flag-render.mjs` as a generator input alongside `v4` (or as `design-tokens/icons/flags/`), add the web/iOS/Android flag generators described above.
- 4. Bump the registry to **4.1.0** (additive ids → minor, per governance), run `npm run icons:v4`, `icons:v4:raster`, `icons:v4:package`, `icons:v4:swift`, `icons:v4:android`, then `npm run icons:validate`.
- 5. Regenerate `v4` catalog/report PDFs; add the CHANGELOG entry; delete the promoted modules from the proposal so it stays a delta.

Open decisions

- **Skeleton glyphs vs CSS skeletons.** Web already draws shimmer blocks in CSS; `placeholder.text/list/card` matter most on iOS/Android and in empty-state illustrations. Keep all six, or keep `avatar/plant/chart` and drop the three pure skeletons?
- **Flag scope.** Starter set of 14. If the picker is language-based rather than region-based, `flag.gb` vs `flag.us` for "English" is a product decision (many apps show neither and use `utility.language + text`).
- **Tier of `plant.pet.*` and `care.misting/rotating`.** Proposed P2 (library) until the species attributes / care types exist in the data model; promote to P1 with authored micro grades when they ship.
- **Animation.** `status.loading.dots` is a static frame; a Lottie bounce (3 × 120 ms stagger) is a natural P0-style addition but out of scope for a P1 status icon under DESIGN.md §7.

Appendix — build evidence

ARTIFACT	PATH
authored geometry	<code>design-tokens/icons/v4.1-proposal/src/*.mjs</code> , <code>flags/src/flags.mjs</code>
compiled registry / manifest	<code>registry.proposal.json</code> (520b57ace477bc33...), <code>index.proposal.json</code>
SVGs	<code>svg/<variant>/<id>@<size>.svg</code> — 816 files
gates	<code>npm run icons:proposal:check (structural)</code> · <code>npm run icons:proposal:raster</code>
sheets	<code>npm run icons:proposal:sheets</code> → <code>reports/v4-sheets/proposal/</code> (gitignored working artifacts)
review package	<code>reports/plantim-icons-v4.1-proposal-catalog.html</code> · <code>...-catalog.pdf</code> · this report